PHASE 2 DOCUMENT

ComicCraft: AI Comic Story Creator using Gemini Models

Phase Name: System Design and Architecture

Project Name	ComicCraft: AI Comic Story Creator using Gemini Models
Phase	Phase 2 – System Design and Architecture
Project Type	Generative AI Comic Creation Application
Main Technology	Google Gemini Models
Prepared By	_____
Guide / Faculty	_____
Date	_____

Academic Project Phase Documentation

1. Phase Overview

This is Phase 2 of **ComicCraft: AI Comic Story Creator using Gemini Models**. Based on the requirements identified in Phase 1, this phase defines the technical design of the application. It describes the architecture, application modules, data design, user roles, interface flow, comic-generation pipeline, and secure backend connection to Gemini models.

2. Design Goals

The ComicCraft design should make comic creation simple for beginners while allowing users to control and edit generated content. The system should be modular, responsive, secure, and easy to maintain. It should separate the frontend interface from backend AI processing so that Gemini API credentials are protected and future features can be added without rebuilding the entire application.

3. High-Level Architecture

ComicCraft follows a layered client-server architecture. The frontend provides user screens for account access, comic creation, story editing, project management, and export. The backend handles authentication, input validation, comic-generation requests, prompt construction, Gemini communication, response processing, and database operations. The database stores users, comic projects, story drafts, characters, scenes, panels, and export history. Gemini models are accessed only through the protected backend integration layer.

4. Architecture Components

Presentation Layer: Responsive web interface for login, dashboard, Create Comic, story settings, character editor, panel editor, preview, saved projects, and export.

Application Layer: Backend services for user management, business rules, input validation, generation control, story editing, version management, and error handling.

Data Layer: Database services for user accounts, comic projects, stories, character profiles, scenes, panels, dialogues, user settings, and audit records.

AI Integration Layer: A secure backend service that converts user choices into structured prompts, communicates with Gemini models, and returns validated comic content.

Security Layer: Authentication, authorization, secure secret storage, input controls, request rate limits, content safety checks, and protected data access.

5. Major Modules

1. Authentication and User Profile Module: Manages registration, login, logout, user preferences, and account settings.

2. Comic Project Management Module: Allows users to create, rename, save, reopen, duplicate, archive, and delete comic projects.

3. Story Idea and Settings Module: Collects the main idea, genre, theme, tone, audience, language, comic length, and style preferences.

4. Gemini Story Generation Module: Generates title ideas, premise, story outline, character descriptions, scene sequence, panel descriptions, dialogues, and captions.

5. Character and Scene Editor: Allows the user to edit character traits, story events, settings, and scene order.

6. Panel Script Editor: Organizes comic pages and panels with visual descriptions, dialogue, narration, captions, and sound effects.

7. Preview and Export Module: Shows a structured comic-script preview and exports content in selected text-based formats.

8. Admin and Safety Module: Manages user access, content policies, reporting, usage limits, and system settings.

6. User Roles and Access

Creators can create and manage their own comic projects, request AI-generated content, edit drafts, save versions, and export their work. Administrators can manage users, system configuration, usage limits, and content-policy controls. If an educator role is included in a future version, educators can guide classroom activities or share templates without accessing private student projects unless users explicitly share them. Role-based access control ensures that each user sees only the functions and records allowed for that role.

7. User Interface Flow

The user journey begins with registration or login. After authentication, the dashboard displays options such as Create New Comic, My Projects, Templates, and Profile. In Create New Comic, the user enters an idea and selects genre, tone, audience, length, and language. The application then displays the generated outline. The user can refine characters and scenes, open the panel editor, edit dialogue and captions, preview the comic script, save a version, and export the final result.

8. Comic Generation Pipeline

1. User enters a story concept and comic preferences.
2. The frontend validates required fields and sends the request to the backend.
3. The backend checks the authenticated user and request limits.
4. The backend creates a structured Gemini prompt containing the selected settings.
5. Gemini generates the requested comic content.
6. The backend parses, validates, and formats the generated output.
7. ComicCraft stores the draft under the user’s project.
8. The frontend displays the outline, scenes, and panel script for editing.
9. The user saves, regenerates selected parts, or exports the completed comic draft.

9. Data Design

The proposed data model includes Users for account and role details; ComicProjects for project metadata; StorySettings for genre, tone, audience, and language choices; Characters for names,

roles, traits, and descriptions; Scenes for story events and locations; ComicPages for page ordering; Panels for panel descriptions and visual instructions; DialogueItems for character speech, narration, captions, and sound effects; DraftVersions for revision history; and ExportRecords for generated files. Data must be associated with the correct authenticated creator account.

10. Secure Gemini Integration Design

Gemini models will be accessed only by the backend. The API key will be stored in protected environment variables or a secure secrets manager, never in browser code or public repositories. The backend provides internal endpoints for story generation and controlled regeneration. It validates the user request, applies prompt templates, enforces content rules and rate limits, sends the request to Gemini, processes the response, and returns structured content to the authenticated frontend.

11. Content Safety and Ethics Design

ComicCraft should guide users toward original and responsible content. The system should reject or safely handle prohibited requests, avoid misleading impersonation, discourage direct reproduction of copyrighted characters or living artists' styles, and clearly label generated material as AI-assisted. The design should allow users to report inappropriate content and should record limited audit information for moderation, while protecting user privacy.

12. Error Handling and Reliability

The application should handle missing input, expired user sessions, network interruptions, unsupported requests, Gemini service errors, quota limits, malformed AI output, and database failures. Users should receive understandable messages such as "Please add a story idea" or "Generation is temporarily unavailable; try again later." The system should preserve the last saved draft so that a failed generation request does not cause a user to lose existing work.

13. Proposed Technology Stack

The frontend can use React, Vue, Angular, or standard HTML/CSS/JavaScript. The backend can use Node.js with Express or Python with FastAPI/Django. Firebase, PostgreSQL, or MongoDB can store project data. Gemini models provide the generative AI capability through the backend. Cloud deployment may use Firebase, Google Cloud, Vercel, or another approved platform. The final stack should be chosen based on the project team's skills and available infrastructure.

14. Phase Deliverables

The deliverables of Phase 2 are the high-level architecture, module descriptions, user-role definition, interface flow, comic-generation pipeline, database entity plan, Gemini integration design, safety and security design, error-handling plan, and development roadmap.

15. Planned Next Phase

The next phase will be **Phase 3: Gemini Model Integration and Core Backend Development**. It will implement the secure backend service, protected API configuration, story-generation endpoints, prompt templates, request validation, response processing, and initial comic-generation functionality.

16. Conclusion

Phase 2 transforms the ComicCraft requirements into a structured application blueprint. The proposed design gives users an intuitive comic-creation workflow while securely connecting the system to Gemini models through the backend. This architecture provides a scalable base for developing the core AI comic-generation features in the next phase.

Phase Approval

Prepared by	_____
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